

Answer Key - Final Exam

Exercise 1 1. Simulation of a r.v X following the Poisson distribution:

$$P(X = k) = p_k = \frac{\lambda^k}{k!} e^{-\lambda}, k \in \mathbb{N} \text{ we can write: } p_{k+1} = \frac{\lambda}{k+1} p_k \text{ with } p_0 = e^{-\lambda}$$

Algorithm:

1) Set $k = 0, p = e^{-\lambda}, F = p;$

2) Generate $u \in \mathcal{U}[0, 1]$.

3) While $u \geq F$ set $p = \frac{\lambda}{k+1} p, F = F + p$ and $k = k + 1$.

(1 pt)

4) Set $X = k$.

Application: for $\lambda = 1$,

1)	$\begin{array}{l} k = 0 \\ p = e^{-1} = 0.3678 \\ F = 0.3678 \\ u = 0.0159 < F \\ X_1 = 0 \end{array}$	(0.25 pt)	2)	$\begin{array}{l} k = 0 \\ p = e^{-1} = 0.3678 \\ F = 0.3678 \\ u = 0.3953 > F \Rightarrow \\ p = \frac{\lambda}{1} \times 1 = 0.3678 \\ F = 0.3678 + 0.3678 = 0.7356 \\ k = 1 \\ u = 0.3953 < F \Rightarrow \\ X_2 = 1 \end{array}$	(0.25 pt)
3)	$\begin{array}{l} k = 0 \\ p = e^{-1} = 0.3678 \\ F = 0.3678 \\ u = 0.8145 > F \Rightarrow \\ p = \frac{\lambda}{1} \times 1 = 0.3678 \\ F = 0.3678 + 0.3678 = 0.7356 \\ k = 1 \\ u = 0.8145 > F \Rightarrow \\ p = \frac{\lambda}{2} \times 0.3678 = 0.1839 \\ F = 0.7356 + 0.1839 = 0.9195 \\ u = 0.8134 < F \\ X_3 = 2 \end{array}$	(0.5 pt)	4)	$\begin{array}{l} k = 0 \\ p = e^{-1} = 0.3678 \\ F = 0.3678 \\ u = 0.8397 > F \Rightarrow \\ p = \frac{\lambda}{1} \times 1 = 0.3678 \\ F = 0.3678 + 0.3678 = 0.7356 \\ k = 1 \\ u = 0.8397 > F \\ p = \frac{\lambda}{2} \times 0.3678 = 0.1839 \\ F = 0.7356 + 0.1839 = 0.9195 \\ u = 0.8397 < F \\ X_4 = 2 \end{array}$	(0.5 pt)

The generated sample is $\{0, 1, 2, 2\}$.

2. Let the random variable X with pdf $f(x) = \begin{cases} \frac{1}{4} & \text{if } 0 < x < 1 \\ x - \frac{3}{4} & \text{if } 1 \leq x \leq 2 \end{cases}$.

- Normalization of f : Let $g(x)$ the pdf of Y , st: $Y \sim \mathcal{U}[0, 2]$, and $g_Y(x) = \mathbb{I}_{[0,2]}(x)$, then

$$\begin{aligned} C &= \max_{[0,2]} \left| \frac{f(x)}{g(x)} \right| = 2f(2) = \frac{5}{4} \\ \Leftrightarrow 0 &\leq \frac{f(x)}{Cg(x)} = \frac{4}{5}f(x) \leq 1 \end{aligned}$$

(0.5 pt)

- Calculate x a value of Y : let $U \in \mathcal{U}[0, 1]$ so: $x = 2u$

Algorithm:

1. Generate $U_1, U_2 \in \mathcal{U}[0, 1]$,

2. Calculate $X = 2U_1$,

3. Calculate $\frac{4}{5}f(x) = \begin{cases} \frac{1}{4} & \text{if } 0 < x < 1 \\ \frac{8}{5}u_1 - \frac{3}{5} & \text{if } 1 \leq x \leq 2 \end{cases}$,

4. If $u_2 \leq \frac{4}{5}f(x)$ accept X , else reject X ,

(1 pt)

5. Returne to 1.

Application

App	u_1	$x = 2u_1$	$\frac{4}{5}f(x)$	u_2	Test	Decision
1	0.0159	$0.0318 \in [0, 1]$	0.25	0.3593	$u_2 > \frac{4}{5}f(x)$	reject $X = 0.0318$
2	0.8145	$1.629 \in [1, 2]$	0.5532	0.8397	$u_2 > \frac{4}{5}f(x)$	reject $X = 1.629$
3	0.3090	$0.618 \in [0, 1]$	0.25	0.8100	$u_2 > \frac{4}{5}f(x)$	reject $X = 0.618$

(1.5 pt)

Exercise 2 1. Estimation of $I = \int_0^{\frac{\pi}{2}} \sin(x) dx$ using Monte Carlo method (average sampling)

a) We can see that

$$I = \int_{\mathbb{R}} \frac{\pi}{2} \sin(x) \times \frac{2}{\pi} \mathbb{I}_{[0, \frac{\pi}{2}]}(x) dx = \mathbb{E} \left[\frac{\pi}{2} \sin(X) \right],$$

where $g(x) = \frac{\pi}{2} \sin(x)$ and $f_X(x) = \frac{2}{\pi} \mathbb{I}_{[0, \frac{\pi}{2}]}(x)$ is pdf of r.v. X following a uniform distribution $\mathcal{U}[0, \frac{\pi}{2}]$. Applying the MCM, we can estimate I by

$$\hat{I}_N = \frac{1}{N} \sum_{i=1}^N \frac{\pi}{2} \sin(X_i),$$

(1 pt)

with $(X_i)_{1 \leq i \leq N}$ i.i.d. following $\mathcal{U}[0, \frac{\pi}{2}]$.

b. Let $Z_i = \frac{\pi}{2} \sin X_i$; given that

$$\begin{aligned} \mathbb{E}[Z^2] &= \mathbb{E} \left[\frac{\pi^2}{4} \sin^2(x) \right] = \frac{\pi^2}{4} \int_0^{\frac{\pi}{2}} \frac{2}{\pi} \sin^2(x) dx \\ &= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \frac{1 - \cos(2x)}{2} dx < +\infty, \end{aligned}$$

then, we can calculate the variance of $\hat{I}_N$.

$$\text{Var}(\hat{I}_N) = \frac{\text{Var}(Z)}{N}.$$

Remak that $I = [-\cos x]_{x=0}^{x=\frac{\pi}{2}} = 1$. On the other hand,

$$\text{Var}(Z) = \mathbb{E}[Z^2] - I^2 = \int_0^{\frac{\pi}{2}} \frac{\pi^2}{4} \left(\frac{1}{2} - \frac{\cos(2x)}{2} \right) \frac{2}{\pi} dx - 1$$

$$\text{Var}(Z) = \frac{\pi}{2} \left[\frac{1}{2}x - \frac{1}{4} \sin(2x) \right]_0^{\frac{\pi}{2}} - 1 = \frac{\pi^2}{8} - 1,$$

(1 pt)

hence

$$\text{Var}(\widehat{I}_N) = \frac{1}{N} \left(\frac{\pi^2}{8} - 1 \right). \quad (1)$$

2. Estimate $I = \int_{\mathbb{R}} g(x) f_X(x) dx = \int_0^{\frac{\pi}{2}} \frac{\pi}{2} \sin(x) \times \frac{2}{\pi} dx$, using the importance sampling method

a) We approximate $\sin(x)$ by an odd polynomial x of degree 1 equal to 0 for $x = 0$ and $\frac{\pi}{2}$ for $x = \frac{\pi}{2}$. So we define $\widetilde{f} \sim |g \times f|$ by $\widetilde{f}(x) = \lambda x \times \frac{2}{\pi} \mathbb{I}_{[0, \frac{\pi}{2}]}(x)$, then:

$$\int \widetilde{f}(x) dx = 1 \iff \frac{2\lambda}{\pi} \int_0^{\frac{\pi}{2}} x dx = 1 \iff \lambda = \frac{8}{\pi^2}$$

hence, $\widetilde{f}(x) = \frac{8}{\pi^2} x \mathbb{I}_{[0, \frac{\pi}{2}]}(x)$ that is pdf of r.v. Y , then

$$\begin{aligned} I &= \int_{\mathbb{R}} \frac{g(x) f(x)}{\widetilde{f}_Y(x)} \widetilde{f}_Y(x) dx = \int_{\mathbb{R}} \frac{\frac{\pi}{2} \sin(x) \times \frac{2}{\pi} \mathbb{I}_{[0, \frac{\pi}{2}]}(x)}{\frac{8}{\pi^2} x} \times \frac{8}{\pi^2} x dx \\ &= \int_0^{\frac{\pi}{2}} \left(\frac{\pi^2 \sin(x)}{8} \right) \frac{8}{\pi^2} x dx = \mathbb{E} \left[\frac{\pi^2 \sin(Y)}{8} \right], \end{aligned}$$

so we can approximate I by

$$I \approx \widehat{M}_N = \frac{1}{N} \sum_{i=1}^n \frac{g(X_i) f(X_i)}{\widetilde{f}(X_i)} = \frac{1}{N} \sum_{i=1}^n \frac{\pi^2 \sin(Y_i)}{8} \frac{1}{y_i}, \quad (1 \text{ pt})$$

with $(Y_i)_{i=1}^n$ i.i.d of pdf $\widetilde{f}$.

b) Let $W = \frac{\pi^2 \sin(Y)}{8Y}$, we have

$$\begin{aligned} \mathbb{E}[W^2] &= \frac{\pi^4}{8^2} \int_0^{\frac{\pi}{2}} \frac{\sin^2(x)}{x^2} \frac{8}{\pi^2} x dx \\ &= \frac{\pi^2}{8} \int_0^{\frac{\pi}{2}} \frac{\sin^2(x)}{x} dx \leq \frac{\pi^2}{8} \int_0^{\frac{\pi}{2}} \frac{x}{x} dx < +\infty, \end{aligned}$$

then, we can calculate the variance of $\widehat{M}_N$.

$$\text{Var}(\widehat{M}_N) = \frac{\text{Var}(W)}{N}.$$

$$\text{Var}(W) = \mathbb{E}[W^2] - \mathbb{E}[W]^2 \leq \frac{\pi^2}{8} \int_0^{\frac{\pi}{2}} \frac{x}{x} dx - 1 \leq \frac{\pi^3}{16} - 1,$$

so,

$$\text{Var}(\widehat{M}_N) = \frac{1}{N} \left(\frac{\pi^3}{16} - 1 \right). \quad (1 \text{ pt}) \quad (2)$$

3.

a) Comparaison of estimators, from (1) and (2) we have

$$N \left[Var \left(\hat{I}_N \right) - Var \left(\hat{M}_N \right) \right] = \frac{\pi^2}{8} - \frac{\pi^3}{16} = \frac{2\pi^2 - \pi^3}{16} < 0 \quad (0.5 \text{ pt})$$

so the importance sampling does not reduce variance then, $\hat{I}_N$ is a best estimator of I .

b) Give an approximate number of iterations required to achieve an accuracy of the previous estimator ± 0.01 .

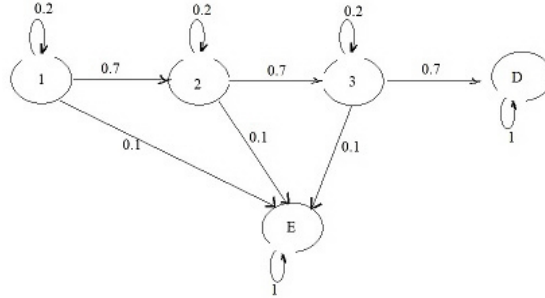
We have $\frac{\sigma}{\sqrt{N}} = \epsilon_N \leq 0.01$ and, since $\sigma^2 = Var \left(\hat{I}_N \right) = \frac{1}{N} \left(\frac{\pi^2}{8} - 1 \right)$ we obtain

$$\begin{aligned} \frac{\sigma^2}{N} &\leq 0.01 \Leftrightarrow \frac{1}{N^2} \left(\frac{\pi^2}{8} - 1 \right) \leq 0.0001 \\ \Leftrightarrow N^2 &\geq \left(\frac{\pi^2}{8} - 1 \right) \times 10^4 = 0.23370055 \times 10^4 \\ \Leftrightarrow N &\geq 48.35 \end{aligned}$$

So, for $N = 49$ we find $Var \left(\hat{I}_N \right) = \frac{1}{49} \left(\frac{\pi^2}{8} - 1 \right) \simeq 0.0048$, acceptable. (1 pt)

Exercise 3 In higher education institution

1. The transition graph associated to the Markov chain with states space $S = \{1, 2, 3, E, D\}$ and, two absorbant state E and D



(0.5 pt)

And its transition matrix

$$P = \begin{pmatrix} 0.2 & 0.7 & 0 & 0.1 & 0 \\ 0 & 0.2 & 0.7 & 0.1 & 0 \\ 0 & 0 & 0.2 & 0.1 & 0.7 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

(0.5 pt)

2. The probabilities that a student obtains his diploma based on his current state in the cursus are:

$$\begin{aligned} p_{1D} &= 0.2p_{1D/11} + 0.7p_{2D} \\ p_{2D} &= 0.2p_{2D/22} + 0.7p_{3D} \\ p_{3D} &= 0.2p_{3D/33} + 0.7p_{DD} \end{aligned}$$

since $p_{DD} = 1$ (absorbant state), the previous system becomes

$$\begin{aligned} p_{3D} &= 0.2p_{3D/33} + 0.7 \\ p_{2D} &= 0.2p_{2D/22} + 0.7 \times 0.2p_{3D/33} + (0.7)^2 \\ p_{1D} &= 0.2p_{1D/11} + 0.7 \times 0.2p_{2D/22} + (0.7)^2 \times 0.2p_{3D/33} + (0.7)^3 \end{aligned}$$

then the solution of this system is

$$\begin{aligned} p_{3D} &= 0.2 \times 0.7 + 0.7 = 0.84, \\ p_{2D} &= 0.2 \times (0.7)^2 + 0.7 \times 0.84 = 0.686, \\ p_{1D} &= 0.2 \times (0.7)^3 + 0.7 \times 0.686 = 0.5488. \end{aligned} \quad (1.5 \text{ pt})$$

3. The average duration of studies: the end of studies is associated with states E and D ; we group them into a state F (finish or end of studies) and then consider the chain with four states.

Let $\overline{T}_i, i \in \{1, 2, 3\}$ the the average stay time in the transient states $\{1, 2, 3\}$ before absorption by F , so the average duration of studies is $\overline{T}_1$ and we have

$$\begin{aligned} \overline{T}_1 &= 1 + 0.2\overline{T}_1 + 0.7\overline{T}_2 \\ \overline{T}_2 &= 1 + 0.2\overline{T}_2 + 0.7\overline{T}_3 \\ \overline{T}_3 &= 1 + 0.2\overline{T}_3 \end{aligned}$$

the system solution is: $\overline{T}_3 = \frac{1}{1-0.2} = \frac{1}{0.8}$ and $\overline{T}_2 = \frac{0.8+0.7}{(0.8)^2} = \frac{1.5}{0.64}$, so:

$$\overline{T}_1 = \frac{1 + 0.7\overline{T}_2}{0.8} = \frac{0.64 + 0.7 \times 1.5}{(0.8)^3} = \frac{1.69}{0.512} = 3.3 \text{ years.} \quad (1.5 \text{ pt})$$

4. Algorithm:

Initial state $X_0 = 1$

- 1) Generate u_1 , if $u_1 < 0.2 \rightarrow X_1 = 1$
if $0.2 \leq u_1 < 0.9 \rightarrow X_1 = 2$
if $0.9 \leq u_1 < 1 \rightarrow X_1 = E$
- 2) If $X_1 = 2$, generate u_2 , if $u_2 < 0.2 \rightarrow X_2 = 2$
if $0.2 \leq u_2 < 0.9 \rightarrow X_2 = 3$
if $0.9 \leq u_2 < 1 \rightarrow X_2 = E$
- 3) If $X_2 = 3$, generate u_3 , if $u_3 < 0.2 \rightarrow X_3 = 3$
if $0.2 \leq u_3 < 0.9 \rightarrow X_3 = D$
if $0.9 \leq u_3 < 1 \rightarrow X_3 = E$
- 4) In step n , if $X_{n+1} = X_n \rightarrow$ return to step n ,
if $X_{n+1} = D$ stop.
if $X_{n+1} = E$ stop.

(1.5 pt)

Exercise 4 In service company

Model A Queueing system $M/M/1$, with arrival rate λ and service rate 2μ .

Since $\frac{\lambda}{2\mu} < 1$, the system is in stationary state, we have:

$$\bar{n} = \frac{\rho}{1 - \rho} = \frac{\lambda}{2\mu \left(1 - \frac{\lambda}{2\mu}\right)} = \frac{\lambda}{2\mu - \lambda} \quad (0.5 \text{ pt})$$

So, from the Little' law the average time spent in the system is

$$\overline{W_{S_A}} = \frac{\bar{n}}{\lambda} = \frac{1}{2\mu - \lambda}. \quad (0.5 \text{ pt}) \quad (3)$$

Model B The sum of two identical systems $M/M/1$, In each one the arrival rate is $\frac{\lambda}{2}$ and the service rate is μ . (0.5 pt)

So, the average time spent in the system is

$$\overline{W_{S_B}} = \frac{1}{2} \frac{1}{\mu - \frac{\lambda}{2}} + \frac{1}{2} \frac{1}{\mu - \frac{\lambda}{2}} = \frac{2}{2\mu - \lambda}. \quad (0.5 \text{ pt}) \quad (4)$$

Model C Queueing system $M/M/2$, with arrival rate λ and service rate is μ per server.

We have $s = 2$,

$$\begin{aligned} p_0 &= \left[1 + \frac{\lambda}{\mu} + \frac{\left(\frac{\lambda}{\mu}\right)^2}{2 \left(1 - \frac{\lambda}{2\mu}\right)} \right]^{-1} = \left[1 + \frac{\lambda}{\mu} + \frac{\lambda^2}{\mu(2\mu - \lambda)} \right]^{-1} \\ &= \left[\frac{2\mu^2 - \lambda\mu + 2\lambda\mu - \lambda^2 + \lambda^2}{\mu(2\mu - \lambda)} \right]^{-1} = \frac{2\mu - \lambda}{2\mu + \lambda}. \end{aligned} \quad (1 \text{ pt})$$

So,

$$\begin{aligned} \overline{W_{S_C}} &= \frac{1}{\mu} + \frac{\left(\frac{\lambda}{\mu}\right)^s}{s\mu s! \left(1 - \frac{\lambda}{s\mu}\right)^2} p_0 = \frac{1}{\mu} + \frac{\left(\frac{\lambda}{\mu}\right)^2}{4\mu \left(\frac{2\mu - \lambda}{2\mu}\right)^2} \times \frac{2\mu - \lambda}{2\mu + \lambda} \\ &= \frac{1}{\mu} + \frac{\lambda^2}{\mu(2\mu - \lambda)^2} \times \frac{2\mu - \lambda}{2\mu + \lambda} = \frac{4\mu^2 - \lambda^2 + \lambda^2}{\mu(2\mu - \lambda)(2\mu + \lambda)} \\ \overline{W_{S_C}} &= \frac{4\mu}{(2\mu - \lambda)(2\mu + \lambda)}. \end{aligned} \quad (1 \text{ pt}) \quad (5)$$

Comparison: Since $\lambda < 2\mu$ then $2\mu + \lambda < 4\mu$, so: $\frac{4\mu}{2\mu + \lambda} > 1$ and $\overline{W_{S_C}} > \overline{W_{S_A}}$. (0.5 pt)

In the other hand, $\overline{W_{S_B}} > \overline{W_{S_A}}$.

From (4) and (5)

$$\frac{4\mu}{(2\mu + \lambda)} < \frac{4\mu}{2\mu} = 2 \text{ so, } \overline{W_{S_B}} > \overline{W_{S_C}}. \quad (0.5 \text{ pt})$$

Then

$$\overline{W_{S_A}} < \overline{W_{S_C}} < \overline{W_{S_B}}.$$

Model A is the best performing. (0.5 pt)